

DENTAL MATERIALS

CHAPTER 2

PHYSICAL PROPERTIES

Y.M

Physical properties

Properties which are not concerned with application of force to the body.

I. Mass related properties:

1) Density (ρ) :

- **Definition:** mass per unit volume.
- **Equation:** $\frac{\text{Mass}}{\text{Vol.}}$
- **Unit:** gm/cm³.

2) Specific gravity:

- **Definition:** ratio of the density of material to the density of water at 4° C.
- **Equation:** $\frac{\text{density of material}}{\text{density of water}}$
 - *density of 1 cubic cm of water at 4°C is 1 ,so specific gravity is equal to density but without units.*
- **Unit:** no unit (unit less).

❖ Importance:

a) retention of upper denture:

1. Light weight upper denture has more retention.
2. Base metal alloys are lighter than gold
3. Non-metallic dentures are lighter than metallic.

b) Light weight complete or partial denture affects comfort of the patient and tissue response.

c) During casting, low density alloy require more casting forces to allow rapid filling of the mold.

II. Thermal properties:

1) Melting & freezing temperature:

- **Definition:** temperature at which materials melts or freeze.

❖ **Importance:** in fabrication of materials outside the mouth:

- a. To determine method of melting and melting machine for rapid effective melting.
- b. Avoid over heating of alloy to avoid volatilization of lower melting metals leading to porosities.
- c. Determine the quality of the investment material that can withstand high temperature without fracture.

2) Heat of fusion:

- **Definition:** amount of heat in calories required to convert 1 gram of a material from solid to liquid at the melting point.
- *N.B: During heating, the temperature of solid increase till it reach melting point then it remains constant till it is completely changed to liquid.*
- **Latent heat of fusion:** On cooling, liberation of the amount of heat gained during liquefaction (change from solid to liquid).
- *N.B: Explain why the temperature remains constant during solidifying liquid; this is due to the liberation of the latent heat of fusion.*
- **Unit:** calories or joule.

❖ **Importance:** During casting alloy, heating must be 100°C above melting temperature to avoid early solidification (in complete casting) by allowing the molten alloy to completely fill the mould before solidification

3) Thermal conductivity (K):

- **Definition:** quantity of heat in calories per second passing through a body of 1 cm thickness and cross section 1 cm^2 , when the temperature difference is 1°C .
- **Unit:** $\text{Cal/sec/cm}^2 / (^\circ\text{C} / \text{cm})$

❖ Importance:

- Disadvantage: Metallic filling material (amalgam or gold) close to the pulp causes thermal irritation to pulp and discomfort to patient during hot and cold food. So, proper amount of dentine is left or cement base is placed.
- Advantages: metallic denture base should conduct heat to keep the underlying tissues in a healthy state by allowing vasodilatation and vasoconstriction.

4) Specific heat or heat capacity (C_p)

- **Definition:** the quantity of heat needed to raise the temperature of 1 gram of substance by 1°C .
- The heat required to raise the temperature of 1 gm of water by 1°C (from 15-16) is 1 calorie so it is taken as a standard.

➤ *N.B: The total heat required depends on:*

- Total mass $\rightarrow$ 100 gm of water requires more calories than 50 gm.*
- Specific heat $\rightarrow$ 100 gm water requires more calories than 100 gm alcohol.*

- Importance:

- During casting, the total amount of heat needed to raise the temperature of an alloy up to melting is determined by its specific heat and total mass.
- Gold alloys have low specific heat so prolonged heating is unnecessary.

c. Thermal diffusivity (Δ):

- **Definition:** -rate at which non-uniform temperature body approach thermal equilibrium.
 - Diffusion rate of heat inside a body.
- **Equation :** $\Delta = \frac{\text{thermal conductivity}}{\text{specific heat} \times \text{density}}$
- **Unit:** mm²/ sec.

❖ **Importance:**

- a. It gives indication of the rate of temp. rise at 1 point when another point reaches high temp.
 - b. Low specific heat and high thermal conductivity of amalgam leads to thermal shock to the pulp.
- *diffusivity and thickness are important factors regarding insulating capacity.*
- *Thermal diffusivity is more important than thermal conductivity when concerned with base.*

d. Coefficient of thermal expansion (α):

- **Definition:** change in length per unit length when temperature increases by 1°C.
 - **Equation :** $\alpha = \frac{(L_{\text{final}} - L_{\text{original}})}{L_{\text{original}} \times (^{\circ}\text{C}_{\text{final}} - ^{\circ}\text{C}_{\text{original}})}$
 - **Unit:** / °C
- *During heating, the energy absorbed causes atoms to vibrate and material expands.*
- *The values are small so expressed as part per million ($10^{-6}/^{\circ}\text{C}$).*

❖ **Importance:**

- a) Coefficient of thermal expansion between the tooth and the restoration should be nearly equal because they expand by heating and contract by cooling, so if there is a great difference it will cause breakage of seal → gaps → marginal leakage → discoloration, recurrent caries
- b) Wax pattern has high coefficient of thermal expansion so it may cause wax distortion which lead to cast distortion.
- c) Thermal expansion of the silica in investment (displacive transformation) will compensate for solidification shrinkage.
- d) Close matching in coefficient of thermal expansion between metal and porcelain in ceramo-metallic restoration will cause bond between them.(metal > porcelain)
- e) Difference in coefficient of thermal expansion between acrylic denture base and artificial porcelain teeth will cause crazing of acrylic.

III. Electrical properties:

1) Electrical conductivity:

- **Definition:** the ability of a material to conduct electric current

➤ N.B: -Resistivity $\propto \frac{1}{\text{conductivity}}$

- Resistivity is more common term

- Unit of resistivity is ohm.cm

❖ **Importance:**

a. According to resistivity of cements :

Zinc oxide eugenol > zinc polycarboxlate > zinc phosphate > glass ionomer

b. Applying electrical stimuli can investigate pain perception threshold.

Cariou teeth are less electrical resistant than sound tooth structure.

IV. other properties:

1) water sorption:

- **Definition:** amount of water adsorbed on the surface or absorbed inside the body.
- **Importance:**
 - a. Acrylic denture base materials tend to water sorption which may lead to dimensional change and warpage. However, Small amount of water sorption compensate for processing shrinkage.
 - b. Hydrocolloid impressions imbibe water if immersed in it so this will lead to dimensional change.(100% humidity is needed to preserve alginate)

2) Fluidity: tendency of liquid to flow.

3) Viscosity: resistance to flow.

- Both are measured by time required for unit amount of liquid to flow through capillary opening of calibrated diameter

❖ Importance:

- a. Fluidity and viscosity play an important role in diffusion and wetting which are important factors in adhesion.
- b. Materials in dentistry are manipulated in fluid and viscous state.
- c. The accuracy of impression depends on its flow.

4) Plasticity:

- indicates that material can be easily and permanently deformation under force.
- related to solids and semi-solids as amalgam and wax

5) Shelf life: deterioration and change in the quality of materials during storage.

V. Optical properties:

1) Nature of light:

- ***Light:*** electromagnetic radiation travels with velocity $2.99 \times 10^8 \text{ m/sec}$.
- Visible electromagnetic radiation ranges from 350-700 nm.
- Wave length below visible: ultraviolet, xrays, gamma and cosmic rays.
- Wave length above visible: infrared, microwave, TV, radio and electric waves.

2) Production of light:

Light is emitted through different sources:

- a. *Thermal:*** when temperature of a body reaches white hot level.
e.g.: The sun and incandescent lamps (tungsten filament is heated)
- b. *Electrical:*** passing electric current through inert gas e.g. Neon
- c. *Chemical:*** some chemicals emit visible light when illuminated.
e.g.: Fluorescence and phosphorescence.

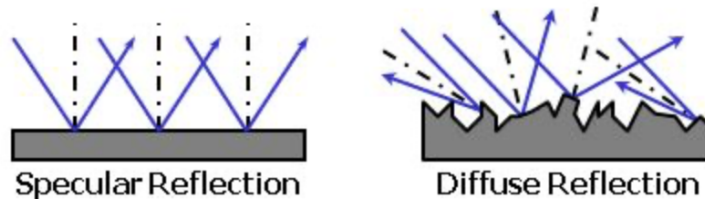
3) Interaction of light with matter:

- When light hits an object the following may occur:

a. Reflection:

When light falls on an object, it is reflected to be seen by our eyes

- ***Specular reflection:*** - smooth surface reflects light in one direction.
 - angle of incidence is equal to angle of reflection.
 - object appears shiny.
- ***Diffuse reflection:*** - rough surface reflects light in all directions.
 - objects appear dull.



❖ **Importance:** Restoration should be smooth and polished to simulate tooth structure.

b. Refraction:

- **Definition:** Change in the direction of light (bending) when entering different medium due to difference of the speed of light in air and other medium.
- $$\text{Refractive index} = \frac{\text{velocity of light in air}}{\text{velocity of light in medium}}$$
- Refractive index of enamel is 1.65

❖ **Importance:**

Control of refractive index between filler (dispersed phase) and matrix in composite and porcelain to have proper translucency.

➤ *N.B: perfect match results in transparent solid, while large difference cause opacity.*

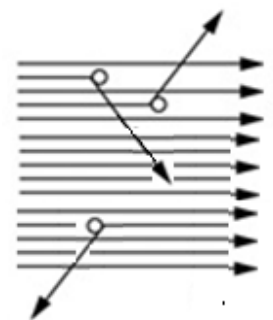
c. Scattering:

- **Definition:** When light passes through a medium which contain inclusions, it will redirected in another direction and it is attenuated (weakened).

❖ **Importance:**

- i. Rough composite scatter more light so they appear opaque and dull.
- ii. Opacifiers added to composite act as scattering centers thus give opaquer shades
- iii. Air bubbles in the restoration act as scattering centers.

N.B: opacity increase with the increase of scattering centers.



d. transmission:

- Complete transmission occurs when all light pass through a medium without attenuation.
- It occurs in transparent materials.

e. Absorption:

- Absorption occurs when the material absorb light by electrons that increases its energy.
 - The intensity of light decrease and will not be able transmitted through the material
 - The more absorption of light, the more opaque of the material.
- *Enamel consist of hydroxyapatite crystal and proteins, when light strikes enamel some is reflected, some refracted, some absorbed and some transmitted.*

➤ ***Properties related to transmission and absorption:***

1) **Transparency:** Passage of light through a material where most of the light is transmitted and some absorbed or reflected.

- The object can be clearly seen through them.
- e.g. Glass and acrylic resin.

2) **Translucency:** Passage of some light through an object and the rest is scattered or reflected.

- The object cannot be clearly seen through them
- e.g.: Enamel, porcelain, composite and pigmented acrylic resin.

3) **Opacity:** No passage of light through an object where all light is absorbed.

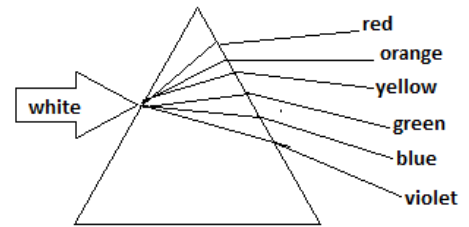
- The object cannot be seen through them.
- e.g.: metals and amalgam.

f. Dispersion:

- When white color pass through a prism it is dispersed to its components these colors are called spectrum of light source,
- The spectrum starts from the shortest wave length which is the violet (350 nm) and ends with the longest wave length which is the red (700 nm).

❖ ***Importance:***

During color matching, incandescent light emits different spectral distribution from neon light.



g. luminescence:

- When the electrons are exposed to light energy, they are activated to a higher energy level and when they return to its normal position, energy is released producing luminescence.
- The emitted light has wave length longer than the exciting waves, so ultra-violet (short wave length, less than 300) may emit visible light (350-700).

fluorescence → immediate emission.

Phosphorescence → delayed emission

❖ ***Importance:***

- i. Human teeth emit blue fluorescence which gives them bright and vital appearance.
- ii. Restorative materials contain fluorescing agents to give natural appearance of tooth structure.

Color

➤ Color vision and perception:

- Light falls on an object to be reflected, transmitted or absorbed.
- The reflected light falls on the eye and converted to nerve impulse to the brain.
- Eye is sensitive to red, green and blue. But most sensitive to green-yellow (550 nm) region and least sensitive to extreme blue and red.

A) Color production by emission (RBG color model):

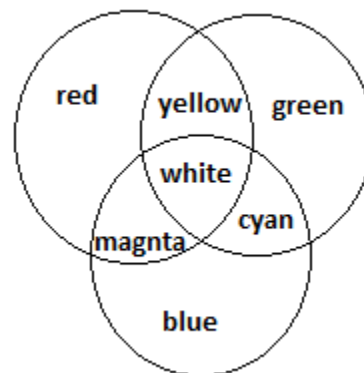
- Television and monitors creates colors by emitting wave lengths that are mixture of red, green and blue (RGB) to stimulate eye
- Therefore produce color spectrum that includes nearly all colors in visible spectrum.

❖ Primary emissive colors RGB:

- They are blue, green and red.
- Mixture of different percentages of primary colors can produce any color within visible range

❖ Additive color mixing:

- The process applied to colored lights.
- Secondary colors result from additive mixing
- Combining the 3 primary colors in equal proportions will result in white light.



B) Production of color by reflection and transmission CMY K color model:

- Printed materials and photographs are considered reflective media, they are visualized by reflection of light on their surfaces
- Slides and transparencies are considered transmissive media, they are visualized by transmission of light through their surfaces.
- These materials are made to absorb some wavelengths and reflect or transmit others to create specific colors

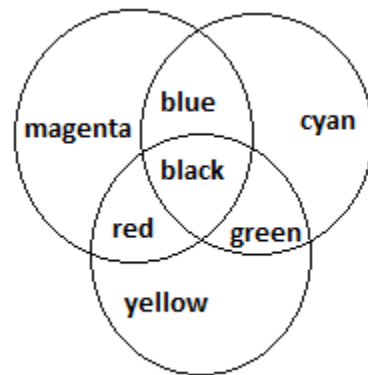
❖ subtractive primary colors CMY:

- The primary color in this model are created by absorption of RGB wavelengths and reflection or transmission of others.

➤ ***Cyan** is produced when red is absorbed while blue and green are reflected*

➤ ***Magenta** is produced when green is absorbed while blue and red are reflected*

➤ ***yellow** is produced when blue is absorbed while red and green are reflected*



- if no wave lengths is absorbed , thus all are reflected and the resultant is white
- Absorption of all RGB wavelengths, the color is black (K)

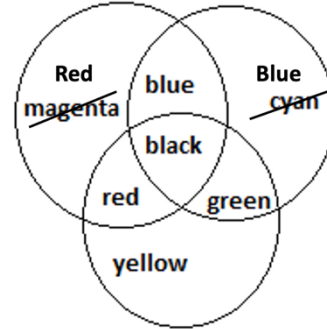
Colors in dentistry (Pigment colors):

Objects contain different pigment colors, they are perceived by transmission or reflection of wavelengths (same as subtractive colors used in color reproduction)

❖ Primary colors (subtractive):

- Similar to subtractive primary colors CMY but they are blue, red and yellow, they are perceived when one of the RGB is absorbed.

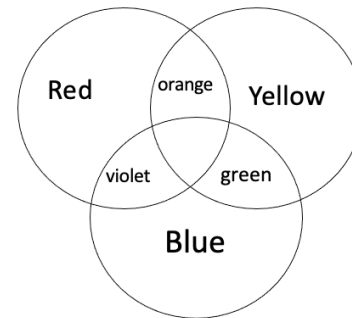
- **Red** → when green is absorbed
- **Yellow** → when blue is absorbed
- **Blue** → when red is absorbed.



❖ Secondary colors:

- Combining 2 primary pigment colors together will result in secondary color.

- red + yellow → **orange**
- yellow + blue → **green**
- blue + red → **violet**



❖ Complementary colors:

- 2 colors are complementary when their mixing results in gray color.

Blue + orange → gray

yellow + violet → gray

Red + Green → gray

➤ Dimensions of color:

1) **Hue (color itself):**

- Describe the pigments of a tooth or restoration (red, blue or yellow).

2) **Chroma (saturation/Purity):**

- Describe the intensity of color, the more wavelengths of color that are reflected, the higher the chroma)

3) **Value (lightness or darkness):**

- It represents the lightness and darkness of color, the more light reflected, the higher the value.

- A black standard have 0 values (low) while white have 10 values (high).
- Value is the most important parameter in dentistry because it is related to the vitality of the teeth.
- Dead tooth have lower value (more dark) while vital tooth have higher value.
- *N.B: if the hue of the restoration matches the tooth but the value is low, the restoration appears non-vital and if the value is too high, the restoration appears too bright (false look)*
- *Adding complementary colors to restoration make it more gray → lower value*

4) Translucency:

- The incisal edge of natural teeth are translucent, thus accurate translucency of restoration cause esthetic success and natural appearance.
- Lab technicians use *densitometer* to measure amount of light transmitted through restoration
- Hue can be easily determined while chroma and value is difficult, thus it is recommended to use 18% gray card as a background when viewing teeth shade to eliminate distraction from surroundings.

❖ Transmitted color:

- When an object transmits a certain color, this is due to absorption of all colors except that color which is transmitted.
- Blue filter transmits blue and absorb all colors

❖ Reflected color:

- An object gains its color by reflecting that color and absorbing all other colors.
- White color reflects all light colors.
- Green colored object absorbs all colors and reflect green.

➤ **Factors affecting color appearance:**

1) source:

- Different sources of light have different color content that daylight differs from incandescent light and fluorescent light.
- Metamerism: objects appear color matched under one light may appear different under other light source
- *color matching and laboratory procedure should be done under 2 or more lightening conditions.*

2) Surroundings:

- Color of wall, lips and cloths of patient modify type of light reaching the object.
- *yellow walls absorb blue and emit more yellow to the illumination.*

3) Object:

- a. Translucency: It controls the lightness and darkness of color
 - *High translucency gives a lighter color appearance (higher value)*
- b. Smooth surface appear shiny and bright than rough surface.
- c. Presence of scattering centers: Increase the opacity and lower the value (more dark).
- d. Fluorescence: It makes human teeth vital as it increase brightness.
- e. Thickness: Thickness affect the appearance of the restoration as increasing thickness will increase opacity, increase chroma and lower the value.
- f. Metamerism: The change in color matching of 2 objects under different light sources
- **Metameric pair** → 2 objects are matched under 1 light source but not matched under other light source.
- **Isomeric pair** → 2 objects are color matched under all light sources.

4) *Observer:*

a. Color vision:

- Some individuals may have color blindness and inability to distinguish certain colors due to absence of cells in the retina

b. Color response:

- Eye responds differently among individuals.

c. Color fatigue: (occurs within 5 sec.)

- Constant stimulus of one color decreases the response to that color.
- A complementary color persists after removal of stimulus.

d. Sex, age may affect color perception.

5) *Color matching:*

- Done using shade guide.

- Recently intra-oral spectrophotometers and colorimeters are used

- Steps for ideal shade selection:

- 1- Teeth should be clean and shade guide should be wet.
- 2- Shade comparison should be made at the start of visit
- 3- Colored clothing should be covered and lipstick removed.
- 4- Patient should be viewed at eye level to use most sensitive part of retina.
- 5- Shade comparison should be done under different light sources (incandescent, fluorescent and day light)
- 6- Shade selection should be quickly
- 7- Eye should be rested by focusing on grey-blue surface as the eye re-sensitize to the yellow color of the tooth.